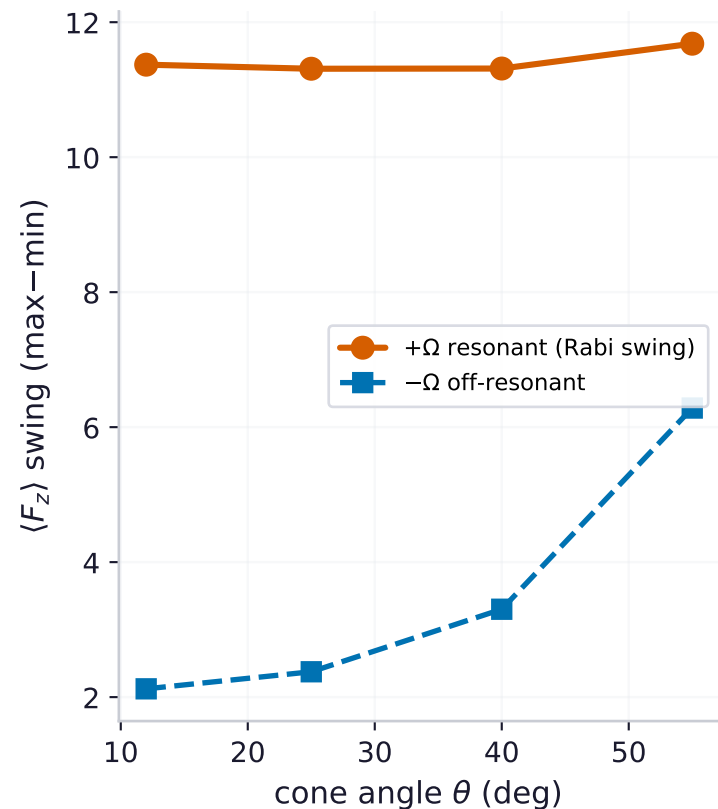
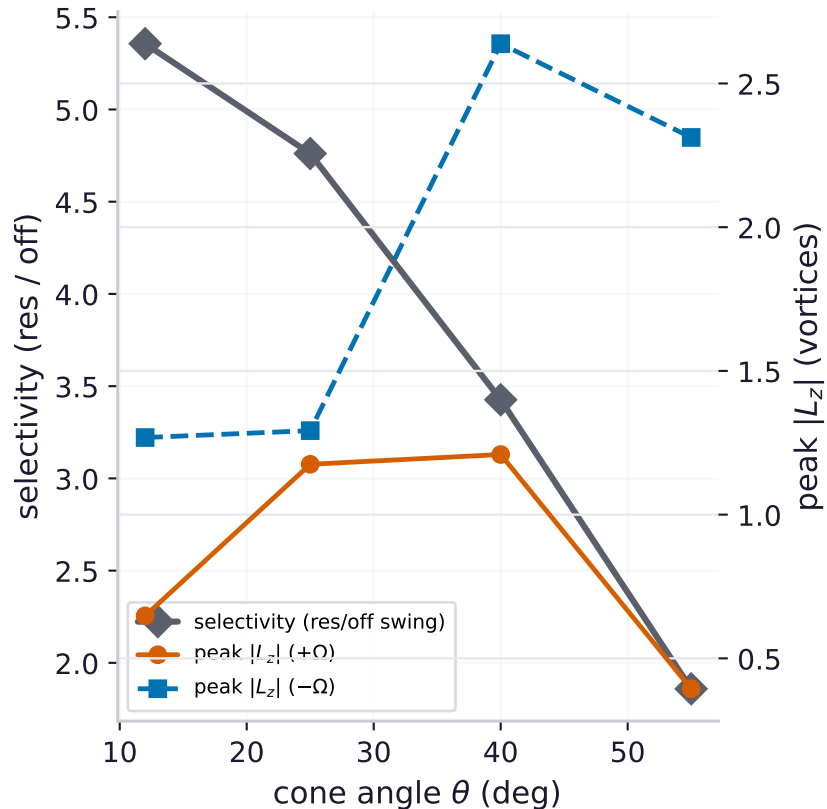


Best field tilt for field-UP chiral excitation (^{151}Eu $F=6$, $m=+F$ metastable, $\gamma B=5.1$)

(a) resonant flip vs off-resonant



(b) one-sidedness $\downarrow$, vortices $\uparrow$ with θ



(c) resonant flip is faster at larger θ

